

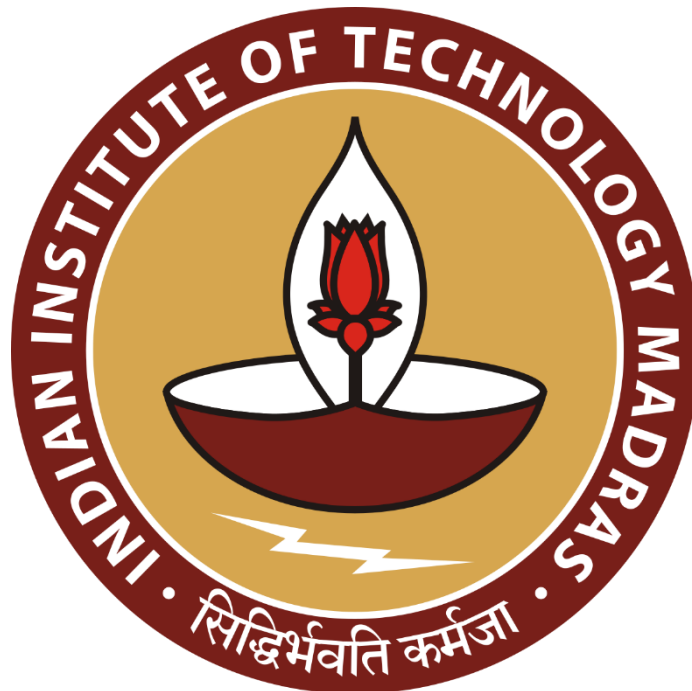
Optimizing Data Management for Efficient Operations in a Soft Drink Wholesale store.

A Proposal report for the BDM Project

Submitted by

Name: KUMAR RISHABH

Roll number: 23f1000391



IITM Online BS Degree Program

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Declaration Statement

I am working on a Project titled “**Optimizing Data Management for Efficient Operations in a Soft Drink Wholesale store**”. I extend my appreciation to **SHIVA ENTERPRISES**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

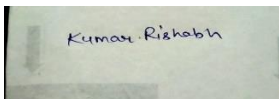
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:

A photograph of a handwritten signature in blue ink on a white background. The signature reads "Kumar Rishabh".

Name: KUMAR RISHABH

Date: 02-02-2024

Table of Contents:

1. Executive Summary and Title	2
2. Organisation Background	3
3. Problem Statement (Listed as objectives)	
3.1 Problem statement 1:	3
3.2 Problem statement 2:	3
4. Background of the Problem	3
5. Problem Solving Approach	4
6. Expected Timeline.....	5
7. Expected Outcome.....	6

1. Executive Summary and Title:

The project focuses on a **Soft Drinks Wholesale** store located at **Yusufpur, Ghazipur, UTTAR PRADESH**. The business is B2B as well as B2C and deals in the segment of drinkable items.

This project aims to analyze sales of a soft drink wholesale store to identify the opportunities and area of improvement and try to provide solutions of it. Specifically, the project will analyze the store's revenue pattern and revenue trends and this will also identify which items and categories are performing low in terms of selling, profit margin and demand. The analysis will be done using Microsoft Excel because it is an organized data of small data set of approximately 3 months. This project will provide detailed information about the project objectives, organization background, methods of data collection, tools to be used for data analysis, and the expected timeline of project completion. The major business issues that the organization is currently facing are related to a very nominal profit due to a large number of inventory stocks, weak operational processes, limited customer engagement, and data maintenance.

By addressing these issues and implementing these potential solutions, **SHIVA ENTERPRISES** can aim to achieve higher profitability, operational efficiency, and enhanced customer satisfactoral lifetime value.

2. Organization Background:

The SHIVA ENTERPRISES is a wholesale dealer of soft drinks and other items situated in Yusufpur, Ghazipur, Uttar Pradesh. This business started in 2018, by Anjani Kr. Gupta. This business is both B2B and B2C category. The store sells the diverse drinkable items from various companies of different category ranges and of different quantity.

With its strategic location, and less competition the store serves in whole area where shopkeepers and local suppliers from different locations and purchase items as per demand made previously by them. The store owner commits to availability of items, and convenience in delivery with the ethos of its marketing and business development. The store is operational full week with 5 days of selling and 2 days of stock adjustment.

Over the years, this store has become more reliable and famous distributor of cold drinks and others items. As the owner seeks to address operational challenges and delivery delay issue sometimes, it remains dedicated to upholding its values while continually evolving to meet the ever-changing needs of its customers.

3. Problem Statement:

- **Problem Statement 1: Inventory Management:** Excessive inventory in stocks is creating **financial blockage, holding huge amount of capital in access stock**, limiting resources which could be invested in business growth. This research endeavors to formulate inventory optimization strategies grounded in data analysis, aiming to minimize surplus stock and enhance capital efficiency for improved business outcomes.
- **Problem Statement 2: Weak Operational Processes:** Weak processes, such as ordering, stock preparation, and customer service, can increase in operational costs and sometime customers have to wait. Inefficient processes lead to prolonged customer wait times and increased operational costs. This study seeks to **identify blockage, smooth workflows, and enhance operational efficiency** to provide a smooth and timely customer experience. By smoothing these processes could not only improve customer satisfaction but also reduce costs and increase the overall efficiency of the store.

4. Background of the Problem:

The store is currently with a notable challenge which has significant implications for its financial and overall business performance. The core issue is inefficient management of stock or we can say inventory, leading to a major capital blockage within the organization. These challenges appear due to a combination of both internal and external factors, which must be thoroughly understood to get fruitful solutions.

- **Major Causes of Problems:** The main cause of this problem is the **lack of demand forecasting** and **stock development**. This has led to excess of stock, resulting in extra resources being engaged in inventory.
- **Internal Problems:** Internally, the store faces various challenges in terms of **process efficiency**. Insufficient **training of staff, lack of digital or computerised data collection**, and a discontinuous approach to stock management have generally obstruct the store ability to smoothly handle stock and meet customer demands punctually.
- **External Problems:** Externally, the store is influenced by various factors like **fluctuation in customer preferences, season wise variations in demand**, and **market trends**. These external factors influence stock whole turnover rates and often lead to overstocking or understocking of certain category of items. Sometimes

other external problems such as weather condition and stock delay by company etc can also lead to weakening in operational process. Inefficient processes lead to prolonged customer wait times and increased operational costs. The combination of these problems has resulted in a terrible situation where financial resources are held in excess stock, leading to reduce profitability. And its consequence is the store ability to establish, increase, and strengthen customer experiences is critically compromised.

5. Problem Solving Approach:

Method of data collection: To collect data from the **Shiva Enterprises** store, at first, I spoke to the owner and informed him about my aim to collect data for a project aimed at improving his store's operations. At first, he seems little bit confused and not provide to give data, but I convinced him by showing previous projects submitted by students through IITM channel, and goodwill IIT Madras has also helped me to get the confidential. The meeting with the owner has been scheduled for **04th of February, 2024**, where I will have the opportunity to discuss target of the project, the data required for this, and the methods of data collection. This first step is very crucial in gaining the support and cooperation of the store owner and his management, putting the base for a successful data collection process.

Analytical Techniques: To constructively address the challenges of inventory management and take advantage of opportunities for enhancing profitability at the Shiva Enterprises store, a mixed problem-solving approach will have to be adopted. This approach involves **use of data-driven modus operandi, advance analytics, and planned interventions to refine inventory, smooth operations, and engagement of customers more effectively.**

1.Demand Prediction and Inventory Enhancement: By Implementing demand prediction techniques for better expectation of customer preferences and enhance in inventory levels. By doing this store owner can prevent overstocking and enhance operational process.

2.Process Automation and Computer Training: By introducing technology and automation wherever possible to smoothing operational processes. Staff training and upskilling in field of excel and tally can also contribute to more efficient and effective service through which owner can make digital and organised data of its business.

3.Profit Analysing: Analyse the profitability and popularity of different category and items. Focus on promoting high-margin items and consider discontinuing items that contribute less in profit and are less in demand.

4.Customer Loyalty Programs: Introduce a loyalty program that rewards repeat customers. This initiative is designed to create a bond with our customers, encouraging them to choose our store consistently over competitors, ultimately boosting customer loyalty and maximizing the overall lifetime value of our store.

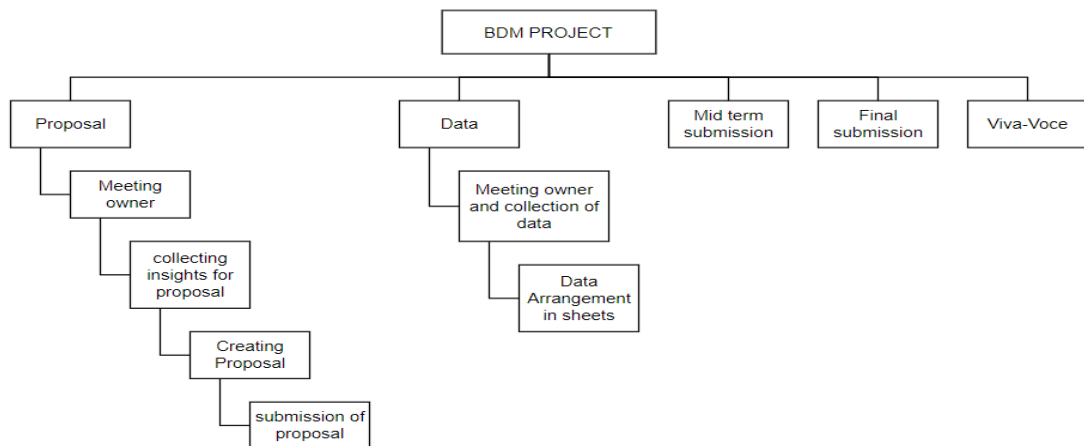
5.Feedback and Improvement: Actively seek feedback from customer and use it to make improvements. Adjust the business strategies based on the changes in market trends and preference of customers.

6.Financial Analysis and Cost Cutting: Conduct a deep financial analysis to identify areas of unnecessary expenditure. Implement cost-cutting measures without compromising.

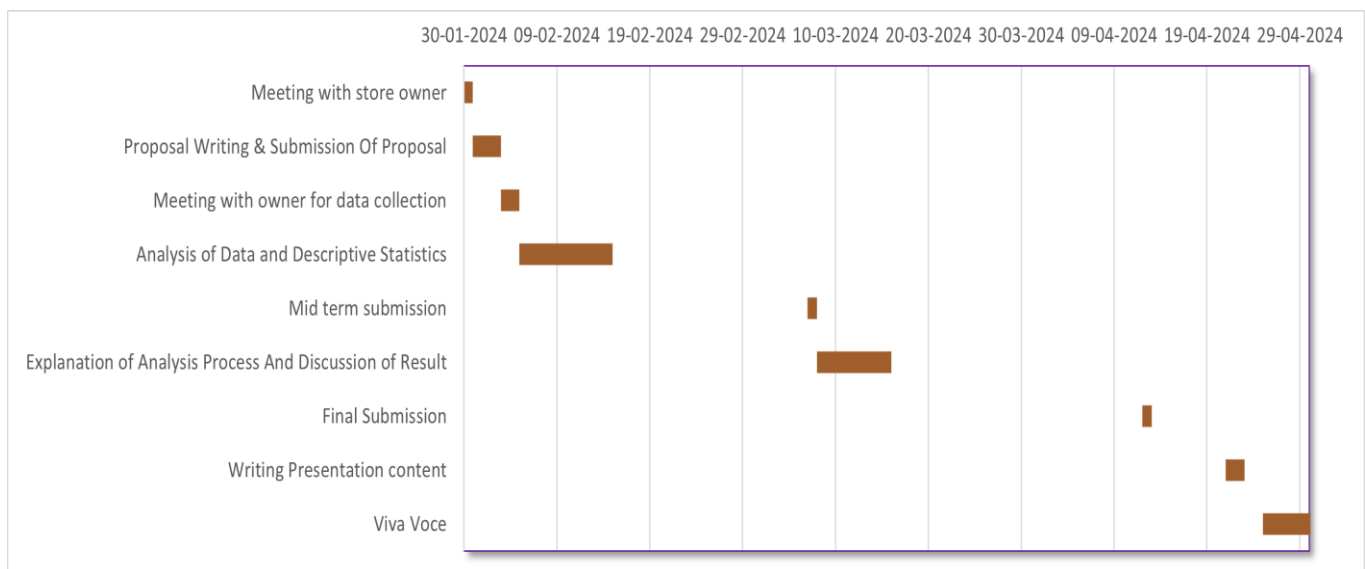
7.Partnerships with Local Shop owners: Build relationships with local suppliers and owners to ensure a steady supply of items to them.

6. Expected Timeline:

a. Work Breakdown Structure:



b. Gantt chart



Sr. No	Activity	Start Date	End Date
1	Meeting with store owner	30-01-2024	30-01-2024
2	Proposal Writing & Submission of Proposal	31-01-2024	03-02-2024
3	Meeting with owner for data collection	04-02-2024	05-02-2024
4	Analysis of Data and Descriptive Statistics	05-02-2024	14-02-2024
5	Mid-term submission	07-03-2024	07-03-2024
6	Explanation of Analysis Process and Discussion of Result	07-03-2024	15-03-2024
7	Final Submission	12-04-2024	12-04-2024
8	Writing Presentation content	21-04-2024	21-04-2024
9	Viva Voce	25-04-2024	30-04-2024

7. Expected Outcomes:

Better Stock Management and Product Insights: By improving stock management and getting better product insights are main aspects of running a successful business. By merging modern technology, data analytics, and tactical planning, **businesses can optimize stock management and gain valuable insights into product performance**, ultimately improving overall efficiency and profitability.

Reduced Returns and Inflate Product Quality: Implementing strategies to reduce returns and enhance product quality leads to a positive change in the business landscape. By giving correct product information, easier return policies, and consistently delivering quality products, customer satisfaction is improved.

Strengthen Financial Performance and viability: Prioritizing viability drive can lead to strengthen in financial performance and long-term viability for business. Secure financial performances and ensuring long-term viability through strategic initiatives which enhance store's performance, improve resource allocation, and fix with market demand.

Data-Driven Decision Culture: Using a data-driven decision culture within an organization is key for success in today's business landscape. By encouraging management to depend on data insights, store can make more **knowledgeable and planned decisions**.